Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

DEVOPSLAB
  |--program8
    |--nodejs-configmap.yaml
    |--nodejs-pod.yaml
    |--server.js

Step 1: Write a Node.js app
  nano server.js

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program8
  GNU nano 7.2                          server.js
const http = require('http');
const port = 3000;

const server = http.createServer((req, res) => {
  res.end('Hello from Node.js running inside Kubernetes!');
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});

                                      [ Read 10 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

Step 2: Create a ConfigMap for the code

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program8
  GNU nano 7.2                     nodejs-configmap.yaml *
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end('Hello from Node.js running inside Kubernetes!');
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create the port yaml using Node.js as a base image

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/program8
GNU nano 7.2                                  nodejs-pod.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs-app
spec:
  containers:
    - name: nodejs-container
      image: node:18-alpine    # lightweight official Node image
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

Step 4: Apply the YAML files

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
```

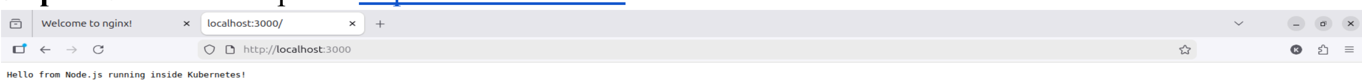
Step 5: Check pod status

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-q5vbf   1/1     Running   0          10m
nginx-deployment-6f9664446b-x84kq   1/1     Running   0          10m
nodejs-pod                          1/1     Running   0          53s
```

Step 6: Access the Node.js app

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl port-forward pod/nodejs-pod 3000:3
000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

Step 7: Check the output in http://localhost:3000



Step 8: Delete all pods

```
^C1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-8xb6z" deleted from default namespace
pod "nginx-deployment-6f9664446b-lp42g" deleted from default namespace
pod "nodejs-pod" deleted from default namespace
```

Step 9: Delete all services

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/program8$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
```